

Number Sense

Answer Key — Worked Solutions

MTH1W · Homework — Answer Key

Worked solutions with method lines. Curriculum: MTH1W (Grade 9 de-streamed math).

1. **[L1 · Integers]** $(-11) + (-8) + (+5) = (-19) + 5 = -14$.
2. **[L1 · BEDMAS]** $2^2=4$; $5 \times 3=15$; $45 - 15 + 4 = 34$.
3. **[L1 · Exponents]** Product rule: $9^{4+3} = 9^7$.
4. **[L1 · Fractions]** $7/10 - 2/10 = 1/2$.
5. **[L1 · Decimals/Percent]** $0.35 = 35/100 = 7/20$; and 35%.
6. **[L1 · Roots]** $\sqrt{144} = 12$; rational ($144 = 12^2$).
7. **[L2 · Fractions]** $3/8 \times 4/9 = 12/72 = 1/6$.
 $1/6 \div 1/3 = 1/6 \times 3 = 1/2$.
8. **[L2 · Exponents]** $(4^2)^2 = 4^4$; $4^4 \div 4^2 = 4^2 = 16$.
9. **[L2 · Percent (word)]** Discount = $0.35 \times 120 = \$42$; sale price = $120 - 42 = \$78$
(or 120×0.65).
10. **[L2 · Ratios (word)]** $350 \div 5 = 70$ g per part; sugar = $2 \times 70 = 140$ g.
11. **[L2 · Rational numbers]** Decimals: $-1/2 = -0.5$, $-2/5 = -0.4$.
Order: $-1/2$, $-2/5$, -0.3 , 0.25 .
12. **[L2 · Roots (estimate)]** $9^2 = 81$ and $10^2 = 100$, and $81 < 85 < 100$.
So $\sqrt{85}$ lies between 9 and 10.
13. **[L3 · Percent (work back)]** $0.75 \times p = 180 \rightarrow p = 180 \div 0.75 = \240 .
Check: $240 \times 0.75 = \$180 \checkmark$.
14. **[L3 · Find the error]** Error: keep the base and ADD exponents; do not multiply the bases.
 $5^2 \times 5^3 = 5^5 = 3125$, not 25^5 .
15. **[L3 · Mixture]** Juice now = $0.30 \times 600 = 180$ mL. Let x = pure juice added.
 $(180 + x)/(600 + x) = 0.40 \rightarrow 180 + x = 240 + 0.4x \rightarrow 0.6x = 60 \rightarrow x = 100$ mL.
16. **[L3 · Rate reasoning]** $(12 + 6)/(1 + 0.5) = 18/1.5 = 12$ km/h.
It divides total distance by total time; the simple mean of 12 and 12 ignores how long each part took.
17. **[L3 · Fractions multi-step]** $3/4 - 1/2 = 1/4$; $2/3 + 1/6 = 5/6$.
 $1/4 \div 5/6 = 1/4 \times 6/5 = 3/10$.

- 18. [L3 · Proportion (scale)]** 2 cm : 15 km, so 1 cm : 7.5 km.
 $9 \times 7.5 = 67.5$ km.